MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY, CHARGE STORAGE

TYPES 7315 AND 7448

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Direct-view charge storage tube, 5-inch, electrostatic focus and deflection, aluminized, P20 phosphor.

DIMENSIONS AND PIN CONNECTIONS: See figures 3 and 4

ABSOLUTE RATINGS:

Viewing gun: See note 1									
Parameter:	Ef	Ec1	Ec2	Ec3	Ec4	Ec5	Ebe	ebe	Escn
Unit:	V	Vdc	Vdc	Vdc	Vdc	Vdc	Vdc	v	kVdc
Maximum:	6.62	0	200	200	300	300	10	20	11
Minimum:	5.98	-200	---	10	---	---	-30	---	---
TEST CONDITIONS:	6.3	Adj (Note 2)	125	Adj (Note 2)	Adj (Note 2)	150	+2	Adj (Note 19)	10

ABSOLUTE RATINGS:

Writing gun: See note 3									
Parameter:	Ef	ehk	Ec1	ed	Ec2, Ec4	Ec3	Scan freq.	Pattern size	Alt
Unit:	V	v	Vdc	vdc	Vdc	Vdc	Hz	---	ft
Maximum:	6.62	±125	0 (Note 26)	500	2,950	1,200	---	---	10,000
Minimum:	5.98	---	-200	---	---	0	---	---	---
TEST CONDITIONS:	6.3	0	Adj	---	2,000	Focus	15,750 x 60 (525 lines) ST	Just overscan	---

GENERAL:

Qualification - Required
Holding period - 96 hours

Ⓑ denotes changes

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METHOD	MEASUREMENT OR TEST	NOTES	GUN	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
---	Vibration	20	-		Width	---	1	mm
1031	Vibration fatigue	-	-		---	---	---	---
---	Shock (1)	4,5	-	15 G; 11 milliseconds; no voltages	---	---	---	---
---	Post-shock (1), vibration, and vibration fatigue test All electrical measurements, QCI tests, parts 1 and 2	22	-		Initial limits as specified			
---	Shock (2)	5,6	-	30 G; 11 milliseconds; no voltages	---	---	---	---
5101	Neck and base alinement (electrostatic types)	-	-		---	---	---	---
5248	Deflection-factor uniformity	-	Write					
1331	Direct-interelectrode capacitance	-	-	Cathode to all				
				Writing gun	C	---	6.0	pF
				Viewing gun	C	---	11.0	pF
				Grid 1 to all				
				Writing gun	C	---	8.0	pF
				Viewing gun	C	---	11.0	pF
				D1 to D2	C	---	2.0	pF
				D3 to D4	C	---	3.0	pF
				D1 to all	C	---	9.0	pF
				D2 to all	C	---	10.0	pF
				D3 to all	C	---	8.0	pF
				D4 to all	C	---	8.0	pF
(B) 1216	Base material insulating quality	-	-		---	---	---	---
1141	Pressure (implosion)	-	-		---	---	---	---
	Quality conformance inspection, part 1	7	-					
1301	Heater current	-	Both		If	0.54	0.66	A
(B) 5401	Voltage breakdown	-	Both		---	---	---	---
---	Grid No. 1 voltage	8	View		vEc1	-60	0	Vdc
---	Grid No. 3 voltage	8	View		vEc3	10	40	Vdc
---	Grid No. 4 voltage	8	View		vEc4	20	90	Vdc
---	Cathode current (max)	-	View	vEc1 = 0; vEc1 = -200 Vdc	vIk	950	6,500	μAdc
---	Erase uniformity ratio	8,9	-		---	---	0.4	---
---	Erasing time	8,10	-		---	---	---	---
---	Grid No. 2 current	8	View	vEc1 = -200 Vdc	vIc2	---	2,500	μAdc
---	Grid No. 3 current	8	View	vEc1 = -200 Vdc	vIc3	---	4,000	μAdc

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METHOD	REQUIREMENT OR TEST	NOTES	GUN	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued	7	-					
---	Grid No. 4 current	8	View	wEc1 = -200 Vdc	vIc4	---	2,500	µAdc
---	Grid No. 5 current	8	View	wEc1 = -200 Vdc	vIc5	---	2,500	µAdc
---	Screen current	8,11	View		Iscn	---	500	µAdc
5221	Light output	8,11	-		---	1,000	---	ftL
(B) 5409	Blemishes, visual output	8,13	-		---	---	---	---
---	Viewing duration	8,14	View		---	20	---	Sec
---	Grid No. 3 current	-	Write	wEc1 = 0; Ebe = -10 Vdc	wIc3	-15	10	µAdc
5251	Grid No. 1 leakage current	8	Write		---	---	---	---
---	Cathode current	8	Write	wEc1 = 0; Ebe = -10 Vdc	wIk	---	3,500	µAdc
5246	Focus voltage (1)	12,13	Write		---	350	750	Vdc
(B) 5402	Grid No. 1 cutoff voltage	8,12, 23	Write	wEc1/cutoff spot	wEco	-54	-92	Vdc
5101	Base alinement (electrostatic types)	8	-	+3D4; pin 4	---	---	---	---
5101	Side terminal alinement (electrostatic types)	8	-	+1D2; trace, screen terminal	---	---	---	---
(B) 5404	Spot position	8	Write	Measured 0.45 inch (11.43 mm) toward screen terminal from tube center	---	---	20	mm
---	Grid drive characteristics (Type 7315)	-	Write	ΔEc1 = wEco -wEc1; wEc1 conditions: screen and backplate floating, vEc2 = vEc3 = 0 Vdc; vEc4 = floating, vEc5 = 25 Vdc; wEc1/Ic5 = 0.1 uAdc	ΔEc1	---	23	Vdc
---	Grid drive characteristics (Type 7448)	-	Write	Same as type 7315, except wEc1/Ic5 = 15 uAdc	ΔEc1	---	50	Vdc
	Quality conformance inspection, part 2							
---	Stray light emission	8,15	Write	wEc2 = wEc4 = 2,950 Vdc; Escn = 11 kVdc	---	---	---	---
---	Line width A (nonstoring) (Type 7315)	8,12 16,23	Write	wEc1 = wEco + ΔEc1	Width	---	0.38	mm
---	Line width A (nonstoring) (Type 7448)	-	Write	Same as type 7315, except wEc1 = wEco + 0.2 ΔEc1	Width	---	0.38	mm
---	Line width B (nonstoring) (Type 7315)	8,12 17	Write	wEc1 = wEco + ΔEc1	Width	---	0.46	mm

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METHOD	REQUIREMENT OR TEST	NOTES	GUN	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
---	Line width B (monitoring) (Type 7448)	-	Write	Same as type 7315, except wEc1 = wEco + 0.2 aEc1	Width	---	0.46	mm
---	Focus voltage (2)	24	Write		---	350	750	Vdc
5231	Spot displacement (leakage)	8	Write	RD1 = RD2 = RD3 = RD4 = 250 K; Rp = 0.25 Meg	Displ	---	10	mm
(B) 5406	Deflection factor	8	Write	+1D2	DF	72	96	Vdc/in.
(B) 5406	Deflection factor	8	Write	+3D4	DF	70	94	Vdc/in.
(B) 5407	Heater-cathode leakage current	-	Write		---	---	---	---
5101	Angle between traces	8	Write		---	88	92	Degrees
1101	Secureness of base, cap, or insert	25	-		---	---	---	---
(B) 1111	Base pin solder depth	-	-		---	---	---	---
(B) 1105	Permanence of marking	-	-		---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	18,21	-	Group D; t = 500 hours (min)	---	---	---	---
---	Life-test end points:							
5221	Light output	8,11	-		---	800	---	ftL
---	Viewing duration	8,14	-		---	15	---	Sec

NOTES:

1. All viewing gun voltages are referred to the viewing gun cathode which is generally operated at ground potential. Grid No. 2 and grid No. 4 of the writing gun and grid No. 2 of the viewing gun are connected together within the tube. Viewing gun circuit values shall be as follows:

   Grid No. 1 circuit resistance- 1.0 maximum megohm
   Series current-limiting resistor (unbypassed) in grid No. 5 circuit - - 0.005 minimum megohm
   Backplate circuit resistance - 0.005 maximum megohm
   Series current-limiting resistance in screen circuit - - - - - - - - - - 1.0 minimum megohm

   Parameters referring to the viewing gun shall have symbol prefixed by "v".

2. Adjust for most uniform, full-size pattern.

3. All writing gun voltages are referred to the writing gun cathode. Grid No. 2 and grid No. 4 of the writing gun and grid No. 2 of the viewing gun are connected together within the tube. Writing gun circuit values shall be as follows:

   Grid No. 1 circuit resistance- 1.0 maximum megohm
   Resistance in any deflecting-electrode circuit (It is recommended that the deflecting-electrode circuit resistances be approximately equal)- 0.1 maximum megohm

   Parameters referring to the writing gun shall have symbol prefixed by "w".

4. There shall be no shorts, momentary or permanent, between the elements during this test.

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NOTES: -Continued

5. The tube shall be subject to 3 shocks in each of the 3 planes, a total of 18 shocks.

   (a) Parallel to the major axis, 3 shocks in each direction.
   (b) Perpendicular to the major axis, 3 shocks in each direction.
   (c) Perpendicular to both the major axis and the plane in (b), 3 shocks in each direction.

6. The tube need not function in any manner after these shocks, and any degree of internal destruction is permissible. There shall be no cracking or destruction of the envelope.

7. The AQL for the combined defectives for attributes in quality conformance inspection, part 1, including inoperatives and mechanical shall be 1 percent.

8. Collimation: With no erase pulse, vary vEc1 to obtain maximum pattern brightness. Adjust erase pulse amplitude to 8 to 10 volts and prf = 1 to 1,000 pps. Adjust pulse width to erase completely in 5 seconds. Adjust vEc4, vEc3, and vEc1 for most uniform erase cutoff of display. (Adjust vEc1 to eliminate flaring of viewing beam at the edge of pattern and vEc4 for most uniform cutoff. Adjust vEc3 for least positive voltage which will provide full size pattern.) The above procedure shall be repeated until most uniform pattern is obtained, maintaining a pattern size of 3.8 inches (96.52 mm) minimum diameter. These voltages shall be maintained for all tests to which this note applies.

9. Erase uniformity ratio shall be determined as follows: With no erasing pulse, overscan the storage surface with writing beam to obtain maximum pattern brightness. Then cut off writing beam. Apply rectangular erasing pulses having an amplitude of between 8 to 10 volts, a pulse repetition rate of 1 to 2 pulses per second, and adjust pulse width to obtain complete erasure in 10 to 15 seconds. Measure time (t_1) or number of pulses from start of erasing to the instant at which any area within the minimum useful viewing diameter is reduced to background-brightness level, and time (t_2) or number of pulses from start of erasing to the instant at which the entire area within the minimum useful viewing diameter area is reduced to background-brightness level. The erase uniformity ratio is defined as $(t_2-t_1)/t_2$.

10. Apply an erase pulse or pulses of 8 to 10 volts amplitude and a total of 60 milliseconds duration. Tubes shall be erased to background brightness at end of the pulse or pulses.

11. With no erase pulse, overscan backplate using 525 line scan with writing beam to obtain maximum pattern brightness. Then vary vEc1 to cut off writing beam.

12. Escn = 1.5 kVdc maximum (adjust for maximum pattern brightness with zero background brightness) and Ebe = -10 Vdc.

13. Quality zones:

    Zone 1: Within a 2.5 inch (63.50 mm) diameter circle concentric with the center of the viewing area.
    Zone 2: The area between 3.8 inch (96.52 mm) and 2.5 inch (63.50 mm) diameter circle concentric with the center of the viewing area.

Dark spots:

The test for dark spots is made at the same conditions as for the light output test.
Definition of dark spots: Areas of the screen which do not brighten during writing.

Allowable number of dark spots:

Zone 1:	Maximum diameter of spot*	Maximum No. of spots
	0.015 to 0.040 inch	2
	0.040 to 0.060 inch	2
Total	0.015 to 0.060 inch	3

Zone 2:	Maximum diameter of spot*	Maximum No. of spots
	0.020 to 0.040 inch	4
	0.040 to 0.060 inch	2
Total	0.020 to 0.060 inch	5

*Dark spots smaller than minimum specified shall not be limited.

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NOTES: -Continued

13. -Continued

Light spots:

The test for light spots is made at the same conditions as for the light output test except that the pattern is erased to viewing beam cutoff by applying erasing pulse to backplate. Definition of light spots: In general there are two types of light spots:

(1) Brilliant spots. Small areas of the screen which are as bright or brighter than maximum brightness of normal screen areas. This type of light spot cannot be erased by normal erasing methods.

(2) Bright spots. Small areas of the screen which are not as bright as the maximum brightness of the normal screen areas. This type of light spot can be eliminated by increasing the amplitude of the erase pulse or by reducing the backplate voltage. Bright spots will be considered objectionable if a reduction in backplate voltage of more than 4.0 volts is required to erase the spot. (With backplate voltage at 5.0 volts and surrounding area of viewing beam cutoff.)

Allowable number of light spots:

Zone 1:	Maximum diameter of spot	Maximum No. of spots
Brilliant spots	---	0
Bright spots	0.030 inch	1

Zone 2:	Maximum diameter of spot	Maximum No. of spots
Brilliant spots	0.020 inch	1
Bright spots	0.030 inch	2

General:

Zone 1: (a) There shall be no more than 2 spots of any type within any 1.5 inch (38.10 mm) diameter circle.
(b) The total number of spots shall not exceed 4.

Zone 2: (a) There shall be no more than 3 spots of any type within any 1.5 inch (38.10 mm) diameter circle.
(b) The total number of spots shall not exceed 6.

14. Viewing duration is the time required for the light output of any unwritten area 1.5 inch (38.10 mm) minimum diameter to reach 10 percent of the maximum light output of that area measured under normal operating conditions.

15. Erase pulses shall be applied such that the tube is erased completely in 10 seconds. With the write gun biased off and no deflection voltages applied, no visible stray emission from the write gun shall be observed.

16. With writing pattern centered with respect to the tube face, the 2940 cycle ST scanning on the vertical deflection shall be adjusted to just overscan. The 60 cycle ST scanning on the horizontal deflection shall be adjusted to produce alternately light and dark lines of approximately equal width. wEc3 shall be adjusted to give clearest, sharpest focus of line at center of tube face. The 60 cycle ST scanning shall then be reduced until pattern lines just overlap at center of the tube face making individual scanning lines indistinguishable.

17. With focus and average deflecting electrode voltage unchanged, the 2940 cycle ST scanning shall be changed to the horizontal deflection and adjusted to just overscan. The 60 cycle ST scanning on the vertical shall be expanded to produce alternately light and dark lines of approximately equal width. The 60 cycle ST scanning shall then be reduced until pattern lines just overlap at center of tube making the individual scanning lines indistinguishable.

18. Life-test conditions are as follows:

Viewing gun: Escn = 10 kVdc, Ebe = +2 Vdc. (An erase pulse is applied to backplate. Erase time is adjusted so that the initial part of the trace written by the writing beam has just erased before re-arrival of the writing beam-time for re-arrival of the writing beam is approximately 9 seconds). Ec5 = 220 to 225 Vdc; Ec1, Ec3, and Ec4 adjusted for brightest, most uniform display. Ec2 = 105 Vdc.

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NOTES: -Continued

18. -Continued

Writing gun: Ec2 = Ec4 = 2,855 Vdc, Ec1 = During horizontal sweep grid No. 1 is pulsed with rectangular pulses of approximately 1.5 to 3.0 µsecond-pulse width. Adjust amplitude of pulses to drive grid No. 1 from cutoff to value necessary to produce maximum brightness of display, but not "blooming." There will be a minimum of 5 pulses per horizontal sweep (i.e., a minimum of 5 range markers).

Radar "B" scan:

Horizontal: Interrupted sawtooth, 500 µseconds off.
Vertical: Triangular sweep, 5 seconds top to bottom, 5 seconds bottom to top.

19. Apply erase pulse to the backplate with amplitude of 8 to 10 volts, prf of 100 to 1,000 pps, and width sufficient to complete erasure in approximately 5 seconds.

20. The tube shall be vibrated in positions X, Y, and Z at an amplitude of 0.040 ±0.0025 inch (1.02 ±0.06 mm), 0.080 ±0.005 inch (2.03 ±0.13 mm) total excursion; from 10 to 50 Hz and back to 10 Hz in 3 to 15 minutes for 15 minutes total at the conditions specified below. A circular trace shall be presented on the screen. The tube shall be vibrated for 60 seconds at the frequency which gives the largest increase in line width or at 50 Hz if there is no change in line width and at any frequency which gives an indication of tube resonance. The maximum increase in line width due to relative motion of tube parts shall be measured and be within the limits specified. Any amount of moving spots are permitted during vibration.

Test conditions: DC voltage referenced to viewing gun cathode.

Escn	10,000 V ± 5%
Ebe	2 V ± 20%
vEc4	See note 8
vEc5	150 V ± 20%
wEf	1,300 V ± 10% #
wEc1	See note 8
wEc3	Focus for best average line width
ED3	100 V ± 10%
ED4	100 V ± 10%
vEc2	100 V ± 10%
vEc1	See note 8
vEk	0 V
ED1	100 V ± 10%
ED2	100 V ± 10%
wEk	1,300 V ± 10% #

wEf must be within 20 V of wEk

21. Initial production lot shall be defined as the first lot submitted for acceptance and that the release of such lot shall be contingent upon satisfactory compliance of one tube. This lot shall represent a quantity of tubes manufactured during a period not to exceed 30 days.

All successive lots shall consist of 25 tubes or less manufactured during a period not exceeding a 3-month interval. Such lots shall be evaluated for life-test acceptance upon satisfactory compliance of life test for one tube per lot.

Satisfactory compliance for three successive lots shall effect early release for all succeeding lots in accordance with MIL-E-1 governing early shipments.

When manufacturing quantities exceed 25 tubes per month, release of such lots shall be contingent upon satisfactory compliance of one tube per month.

22. The post environmental tests for vibration, shock 1 and fatigue shall be made after the completion of all tests requiring post-environmental testing.

23. Adjust average voltage of the deflecting electrodes (within 30 volts of vEc2) to produce the most circular focused spot.

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NOTES: -Continued

24. Escn = 1.5 kVdc maximum (adjust for maximum pattern brightness with zero background brightness). Connect backing electrode (be) to view grid No. 5 (vG5). With slightly overscanned 49 line pattern and with the writing gun defocused (vEc3 = vEk), change vEc1 to zero and adjust vEc3 for best focus at center of tube. Make adjustments as rapidly as possible and, when test is complete, bias off write gun.

25. Test may be made on electrical rejects.

26. Permanent damage to the storage grid may be caused by excess writing current beam density at the storage grid. The maximum grid drive that may be applied to the writing guns depends primarily on the characteristics of the gun. Additional factors related to maximum grid drive are:

    (a) The speed at which the spot is deflected across the storage grid.
    (b) The number of times an element is written.
    (c) The erase rate.
    (d) The operating voltages.

    The curves shown on figures 1 and 2 show the recommended maximum grid drive voltage which may be applied under various conditions of scanning.

    The curves apply to operation at typical operating voltages and with a focused writing beam. It should be noted that these curves are not recommended for optimum operation of the tubes, but are shown as maximum limits within which the tube can be operated without damage.

    (1) The following refers to curve A as shown on figure 1:

    This curve is based on the application of a single grid drive pulse within the interval required for complete erasure. The writing beam is focused and undeflected. If more than one grid drive pulse is applied during the erase interval, the effective pulse duration is the sum of the pulse durations applied during the erase interval. For erase intervals less than 2 seconds, use as the maximum effective pulse duration the sum of the pulse durations in 2 seconds.

    Erasure may be accomplished by a chain of erase pulses, producing gradual erasure, or by a single erase pulse of sufficient duration to erase completely with one pulse ("single shot" erase). With "single shot" erasure, the erase interval is the time between erase pulses.

    Example: Grid drive pulses of 10 microseconds are applied at the rate of 2 pulses per second with erasure adjusted to produce complete erasure in 10 seconds. The effective pulse duration is 200 microseconds.

    (2) The following refers to curve B as shown on figure 2:

    This curve is based on writing a single scanned line, or a single frame of a pattern having no overlapping of the lines. If the line or pattern is written repeatedly in the same location, the equivalent pulse duration determined from curve B should be multiplied by the number of lines or pattern frames written during the interval required for complete erasure. If there is overlapping of lines in the pattern, the equivalent pulse duration determined from curve B should be multiplied by the maximum number of lines written in a 0.020 inch (0.51 mm) increment of the storage grid. The equivalent pulse duration determined from curve B is multiplied by the factors due to repeated writings and overlapping lines. This figure is the effective pulse duration and should be inserted in curve A to determine the maximum grid drive voltage.

    Example: A 500 line pattern of equally spaced lines, adjusted for a 2 x 2 inch (50.80 x 50.80 mm) size, is written 20 times per second with erasure adjusted to produce complete erasure in 10 seconds. Due to repeated writing, the equivalent pulse duration should be multiplied by 200, (20x10). Due to overlapping lines, the equivalent pulse duration should be multiplied by 5, $(500 \div \frac{2.0}{0.02})$. The effective pulse duration is then 1,000 times, (5x200) the equivalent pulse duration determined from curve B.

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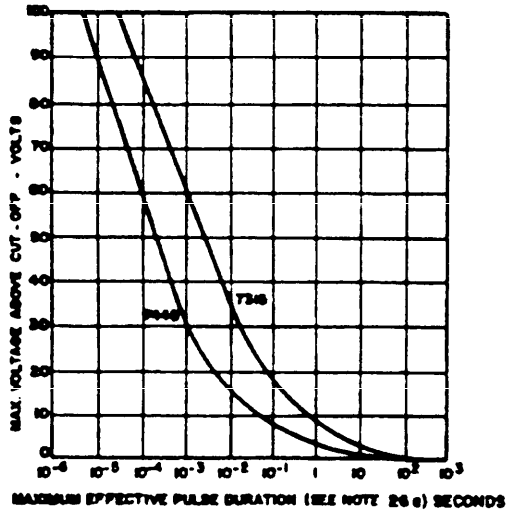


FIGURE 1. Curve A undeflected spot.

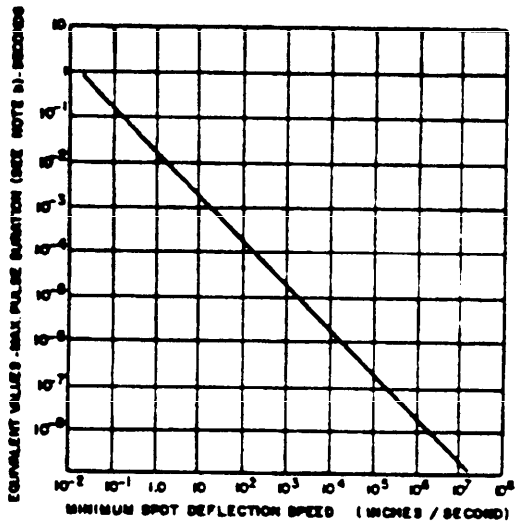


FIGURE 2. Curve B deflected spot.

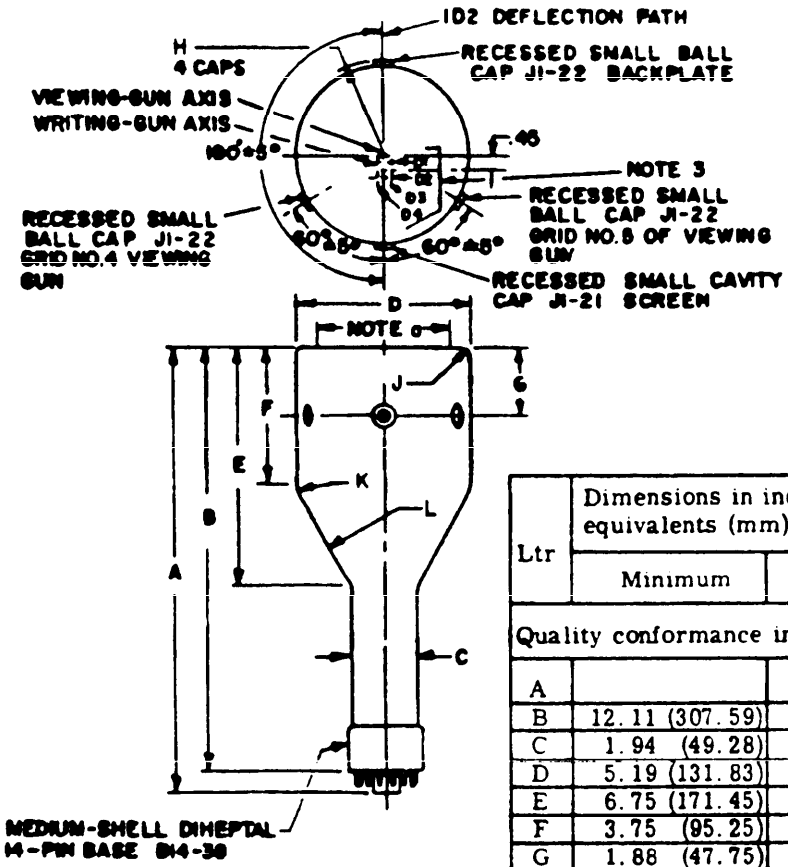
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

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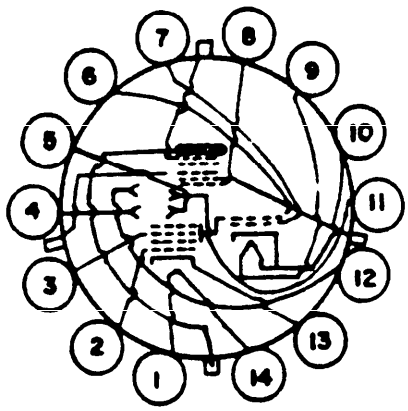


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		13.64 (346.46)
B	12.11 (307.59)	12.89 (327.41)
C	1.94 (49.28)	2.06 (52.32)
D	5.19 (131.83)	5.31 (134.87)
E	6.75 (171.45)	7.25 (184.15)
F	3.75 (95.25)	4.25 (107.95)
G	1.88 (47.75)	2.12 (53.85)
H		2.69 (68.33) R
Reference dimensions (see note c)		
J	.218	(5.54) R
K	1.50	(38.10) R
L	13.00	(330.20) R

NOTES:
a. Minimum useful viewing area may be eccentric with respect to the tube face. The minimum useful viewing area will have diameter of 3.80 inches (96.52 mm).
b. Deflecting electrodes D1 and D2 are nearer the screen; deflecting electrodes D3 and D4 are nearer the base. With D1 positive with respect to D2, the spot will be deflected toward pin no. 8; likewise, with D3 positive with respect to D4, the spot will be deflected toward pin no. 4.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 3. Outline drawing of electron tube types 7315 and 7448.

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Pin No:	1	2	3	4	5	6	7	8	9	10	11
Element:	wH	wG1	wG3	D3	D4	vG2 wG2+4	vG1	vG3	vH	vH	D1
Pin No:	12	13	14	Cap (over pin 3)		Cap (over pin 12)		Cap (on side of tube opposite base key)		Cap (over base key)	
Element:	D2	wK	wH	vG5		vG4		Be		screen	

FIGURE 4. Basing diagram and pin connections of tube types 7315 and 7448.

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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